



Extrapolating a sequence

1 Which of these are triangular numbers? Tick **3** correct answers

☒ 1☐ 8☐ 12☒ 15☒ 21

2 Which of these are prime numbers? Tick **3** correct answers

☐ 1☒ 2☒ 11☒ 13☐ 15

3 7, 8, 15, 23, 38 ... are the first five terms of a sequence. Which of these could be a rule for the sequence? Tick **1** correct answer

☐ Add 1, add 3, add 5 and so on, differences increasing by 2 each time.☐ Add 1, add 7, add 13 and so on, differences increasing by 6 each time.☒ Add the two previous terms to get the next term.☐ Multiply the two previous terms to get the next term.☐ Multiply the previous term by 2 and subtract 1 to get the next term.

4 A linear sequence starts -4, 6, The next term is 16. Fill in the blank



5 A geometric sequence starts $-4, 6, \dots$. What is the next term? Tick **1** correct answer

☐ -12

☒ -9

☐ 4

☐ 9

☐ 12

6 Which of these could be the first 5 terms in a sequence with a common second difference? Tick **2** correct answers

☐ $27, 64, 125, 216, 343, \dots$

☒ $28, 36, 45, 55, 66, \dots$

☐ $30, 52, 76, 98, 122, \dots$

☐ $32, 48, 72, 108, 162, \dots$

☒ $36, 49, 64, 81, 100, \dots$

